

Birmingham Transport Data Analysis

Two GIS case studies: active travel infrastructure mapping and road traffic volume trends

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Abstract

This report presents two independent geospatial analyses of transport conditions in Birmingham, UK, undertaken as applied methods work in support of a transition into transport planning and GIS analysis. Study One maps 46km of cycling and walking infrastructure from OpenStreetMap data and screens the network for candidate discontinuities. Study Two analyses 25 years (2000–2024) of Department for Transport road traffic counts across 591 locations in Birmingham, showing that motor vehicle traffic in 2024 remained approximately 15% below 2019 levels — a substantially larger shortfall than the national picture. Both studies are exploratory and screening-level; neither constitutes validated policy analysis, and both are explicit about the limits of the underlying data.

1. Introduction and Context

Local authorities and consultancies increasingly rely on open geospatial and government administrative data to supplement costly primary data collection. Two data ecosystems are used here: OpenStreetMap (OSM), a volunteered geographic information (VGI) project in continuous use since 2004 (Haklay & Weber, 2008), and the Department for Transport's road traffic count programme, which combines manual counts, automatic counters, and modelled estimates to produce Annual Average Daily Flow (AADF) figures for the road network (Department for Transport, 2023).

VGI datasets such as OSM have been shown to offer good coverage of physical infrastructure in well-mapped urban areas, but their completeness and attribute quality vary by feature type and geography, and this variability has direct consequences when such data feeds into travel demand or infrastructure planning work (Goodchild & Li, 2012; Klinkhardt et al., 2021). This report treats that limitation as a first-class methodological concern rather than a footnote, and designs both studies to make that uncertainty visible rather than to paper over it.

2. Study One: Active Travel Network Mapping

2.1 Method

Cycling and walking infrastructure within central Birmingham was extracted from OSM via the Overpass API, using tag combinations for segregated cycleways, on-road cycle lanes, shared-use paths, and designated footways. Segment lengths were computed using the haversine great-circle formula. A network continuity screen flagged segment endpoints more than 40 metres from the nearest endpoint of another mapped segment, after excluding points near the study-area boundary to avoid edge artefacts.

2.2 Findings

Metric	Value
Study area	Central Birmingham, UK (~5km × 3.3km)
Total mapped active travel network	46.0 km
Segregated cycle paths	39.5 km (85.9%)
Candidate network discontinuities (>40m)	91 of 1,076 endpoints (8.5%)

Segregated provision dominates the mapped network, concentrated along canal towpaths and the inner ring road — consistent with Birmingham's historical reliance on off-carriageway canal corridors rather than on-street cycle lanes. Ninety-one candidate discontinuities were identified; consistent with the VGI quality literature (Goodchild & Li, 2012), these are read here as a triage shortlist for field verification, not as confirmed infrastructure gaps, since many likely represent transitions onto ordinary streets outside the query scope rather than genuine breaks in provision.

3. Study Two: Road Traffic Volume Trends, 2000–2024

3.1 Method

AADF data for all 591 DfT count points within the Birmingham local authority boundary were obtained from the Department for Transport's road traffic statistics platform (Department for Transport, 2023) and aggregated by year and by road. Annual totals were compared against the 2019 pre-pandemic baseline, and against national DfT figures for context.

3.2 Findings

Metric	Value
Study area	Birmingham local authority, UK
Count points analysed	591, 2000–2024
Total motor vehicle traffic, 2024	4.61 million vehicles/day (sum across count points)
Change vs. 2019 (pre-pandemic)	–15.1%
National car traffic vs. 2019 (DfT, 2024)	–2.6%

Nationally, DfT's 2024 headline statistics show car traffic in Great Britain still running 2.6% below 2019 levels, with van traffic exceeding pre-pandemic volumes and HGV traffic down slightly (RAC Drive, 2025; Department for Transport, 2023). The Birmingham count-point aggregate shows a substantially larger shortfall of roughly 15%. This divergence between a large city and the national average is consistent with wider National Travel Survey findings that recovery in trip-making has been uneven across travel purposes and geographies since the pandemic (Department for Transport, 2024a), and is worth investigating further rather than treated as a precise city-level estimate, given the caveats in Section 4.

4. Limitations

- OSM coverage reflects volunteer mapping activity; absence of a feature does not confirm its absence on the ground (Haklay & Weber, 2008; Klinkhardt et al., 2021).
- The 40m discontinuity threshold is a screening heuristic, not a validated design standard, and does not distinguish a genuine gap from a safe transition onto an ordinary street.
- DfT link-level traffic estimates are explicitly described by the department as less robust than regional or national aggregates, since not every count point is re-surveyed every year (Department for Transport, 2023).
- Summing count-point AADF values approximates but does not equal total city-wide vehicle miles, since count points are not evenly distributed across the road network.
- Neither study has been field-validated or peer-reviewed; both should be read as exploratory, screening-level analysis.

5. Conclusions and Recommended Next Steps

- Cross-reference the 91 candidate active-travel discontinuities against Birmingham City Council's own cycling infrastructure records before using them to inform investment decisions.
- Extend the traffic analysis to compare recovery trajectories across West Midlands local authorities, to test whether Birmingham's shortfall is a city-specific or regional pattern.
- Layer Index of Multiple Deprivation and school/employment location data onto both datasets to support equity-focused prioritisation.
- Treat both outputs as a starting point for a scoping study, not as a substitute for primary data collection or statutory consultation.

References

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Data and Licensing

OSM data used in Study One is © OpenStreetMap contributors, available under the Open Database Licence (ODbL). DfT data used in Study Two is available under the Open Government Licence v3.0. All analysis, code, and this report are original work.